

Course

« *Deep Learning* »

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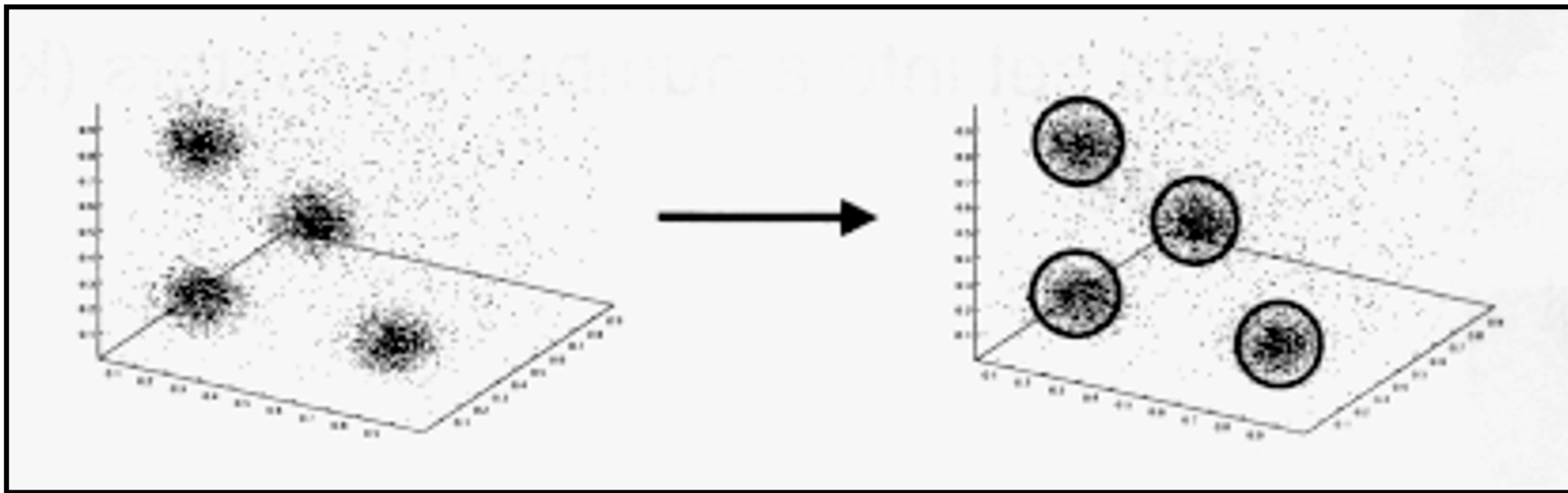
ENSIA

2024-2025

3. *Clustering (unsupervised classification)*

Objective

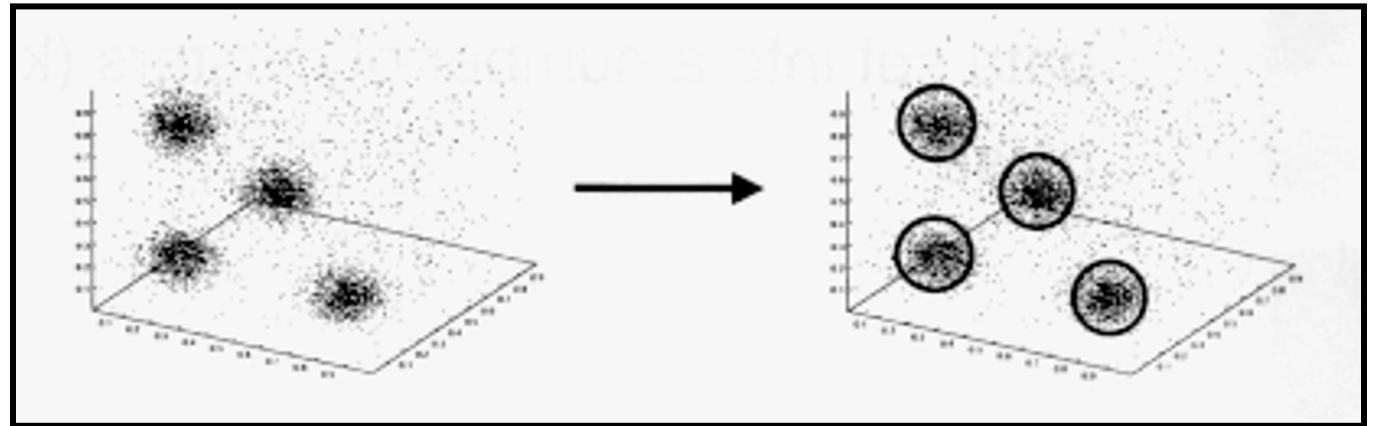
- Automatically partition a dataset using a similarity measure/distance.



3. Clustering (unsupervised classification)

Objective

- Input:
 - A set of N vectors describing N individuals (also called *samples*, *observations...*).
 - A set of parameters (related to the clustering algorithm).
- Output:
 - k clusters
 - + noise (possibly).



3. *Clustering (unsupervised classification)*

Applications

- Summarize large datasets
 - A set of individuals described using d variables (features).

Objective

- Put together similar individuals
- Instead of studying all of the individuals:
 - ⇒ Simply studying the characteristics of the group rather than studying each of the individuals who compose it.
 - ⇒ Study the average individual in each group.

3. Clustering (unsupervised classification)

Applications

- Image segmentation

Input:

- Pixels of an image (values, position...).

Sortie :

Groups of similar and connected pixels. *Homogeneous region = cluster of pixels*



3. *Clustering (unsupervised classification)*

Applications

- **Bioinformatics:** grouping animals according to their biological features to reconstruct phylogeny trees
- **Business:** dividing customers into segments or forming a hierarchy of employees based on salary.
- **Image processing:** grouping handwritten characters in text recognition based on the similarity of the character shapes.
- **Information retrieval:** categorizing search results based on the query.
- ...

3. *Clustering (unsupervised classification)*

In practice: 4 steps

1. Description of the individuals
 - Feature extraction of individuals.
 - Feature selection with respect to the application.
2. Definition of a similarity measure between individuals (a distance, for example).
3. Grouping individuals using their feature vectors and the associated similarity measure.
4. Making use of clusters: Interpretation, abstraction...

3. *Clustering (unsupervised classification)*

Existing techniques: An overview

Data partitioning	Hierarchical	Density-based	Space partitioning
• K-means • CLARANS • Kohonen maps	• Traditional methods • BIRCH	• DBSCAN • DENCLUE	• STING • Wave Cluster • Opti-Grid
• Iterative algorithms that refines an initial clustering.	• Hierarchical partitioning of the data.	• Based on density estimation methods (parametric and non-parametric)	• Space quantization.

3. Clustering (unsupervised classification)

Quality criterion

- The spread of each cluster.

⇒ intra-cluster variance

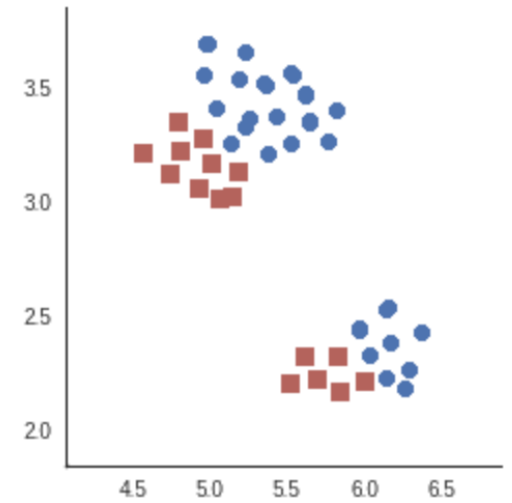
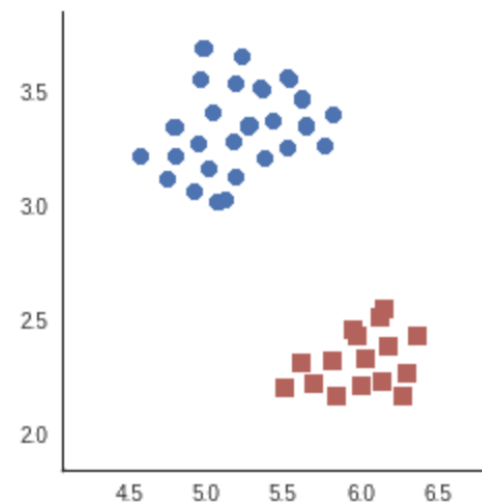
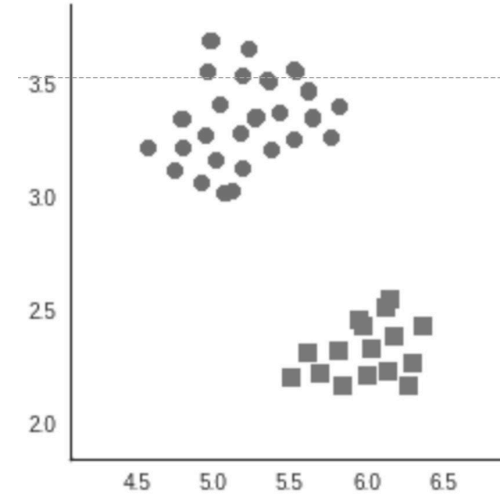
n : total number of individuals (v).

k : number of clusters.

n_j : number of individuals in clusters j

g_j : centroid of cluster j

$$\text{Intra cluster variance} = \frac{1}{n} \sum_{j=1}^k \sum_{i=1}^{n_j} d(v_i, g_j)^2$$



3. Clustering (unsupervised classification)

Quality criterion

- Spacing of clusters from each other.
- If we consider only centroids:
 - ⇒ Study the spread of centroids.
 - ⇒ Inter-cluster variance.

$$\text{Inter cluster variance} = \sum_{j=1}^k \frac{n_j}{n} d(g, g_j)^2$$

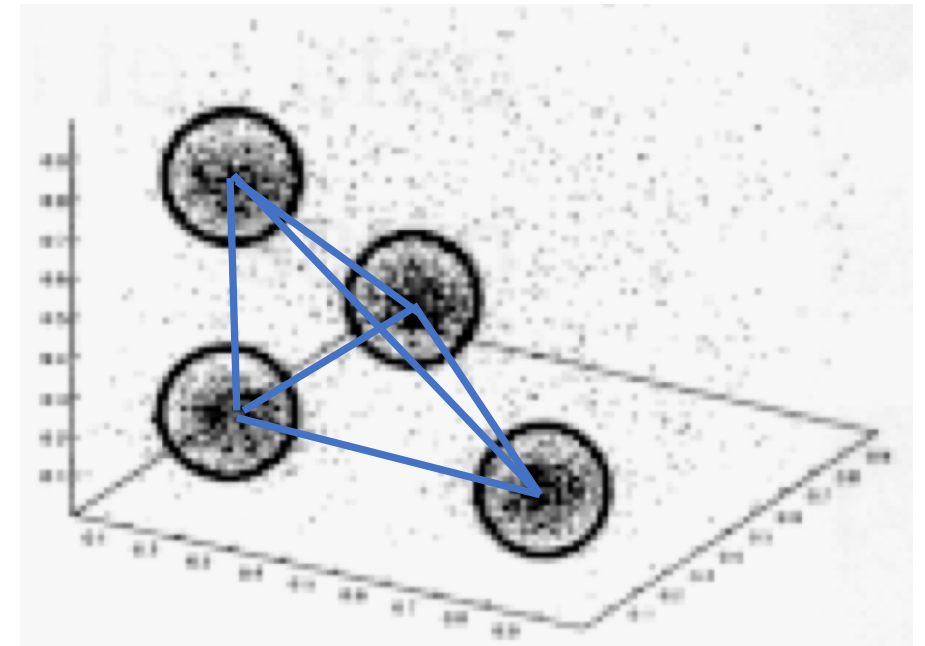
n : total number of individuals.

k : number of clusters.

n_j : number of individuals in clusters j

g_j : centroid of cluster j

g : centroid of the dataset.



3. *Clustering (unsupervised classification)*

Quality criterion

- We seek a clustering in which clusters are:
 1. Tightened on themselves => intra-cluster variance low.
 2. Far from each other => inter-cluster variance high.
- The goal is to minimize the intra-cluster variance and to maximize inter-cluster variance.
- However, the two objectives are equivalent => All you have to do is to optimize one of the two criteria.
- **Total variance = intra-cluster variance + inter-cluster variance**
- The total variance is constant.

(Theorem of Huygens).

3. Clustering (unsupervised classification)

K-Means: The algorithm

Input:

N individuals.

k centers [*randomly chosen at the beginning*].

k empty clusters.

Algorithm:

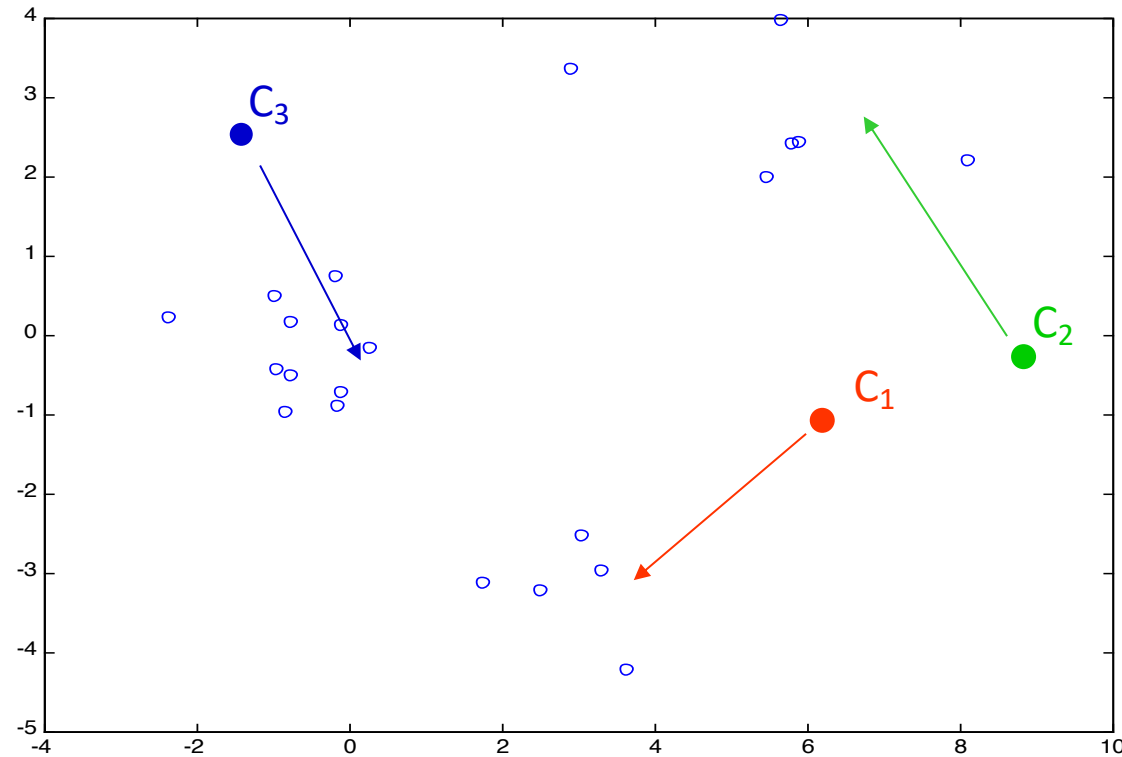
Repeat:

- Distribute the N individuals over the k clusters using the rule of « the nearest center ».
- Compute the new k centers.

Until two consecutive “clusterings” are identical

3. *Clustering (unsupervised classification)*

K-Means: “perform clustering around moving centers”



3. *Clustering (unsupervised classification)*

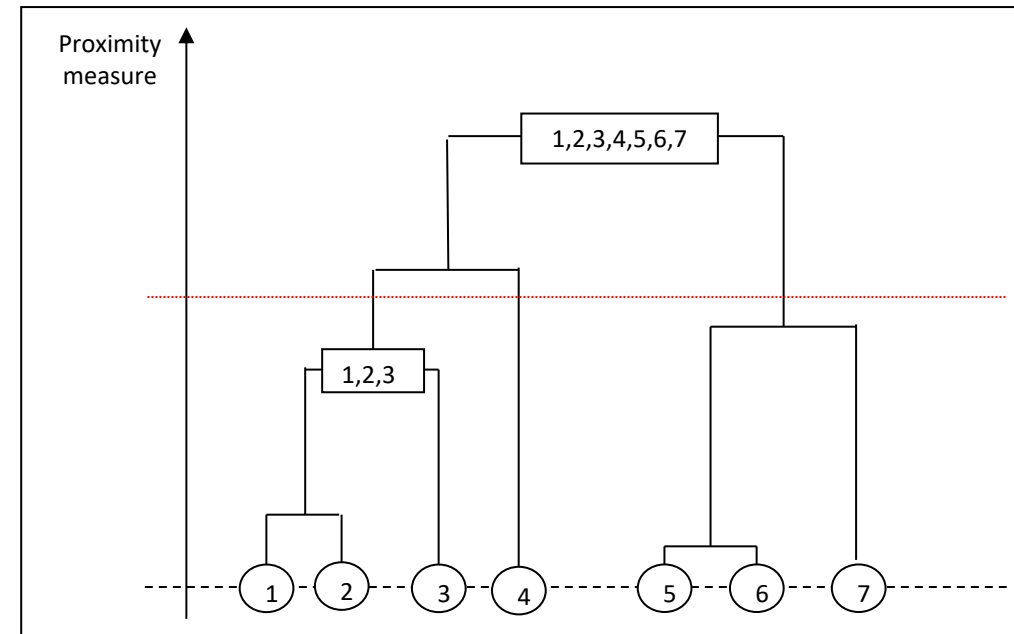
K-Means: The algorithm

- Parameters: The number of clusters
- The algorithm is sensitive to the initial conditions (non-deterministic).
- Iterative algorithm that requires several passes over the dataset.
- Complexity: $O(N)$.

3. Clustering (unsupervised classification)

Hierarchical clustering

- Clusters are visually represented in a hierarchical tree called a **dendrogram**.
- There is no need to pre-specify the number of clusters. Instead, the dendrogram can be cut at the appropriate level to obtain the desired number of clusters.
- Data is easily summarized/organized into a hierarchy using dendrograms. Dendrograms make it easy to examine and interpret clusters.
- Requires a proximity measure between clusters.



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Hierarchical clustering

There are two main types of hierarchical clustering:

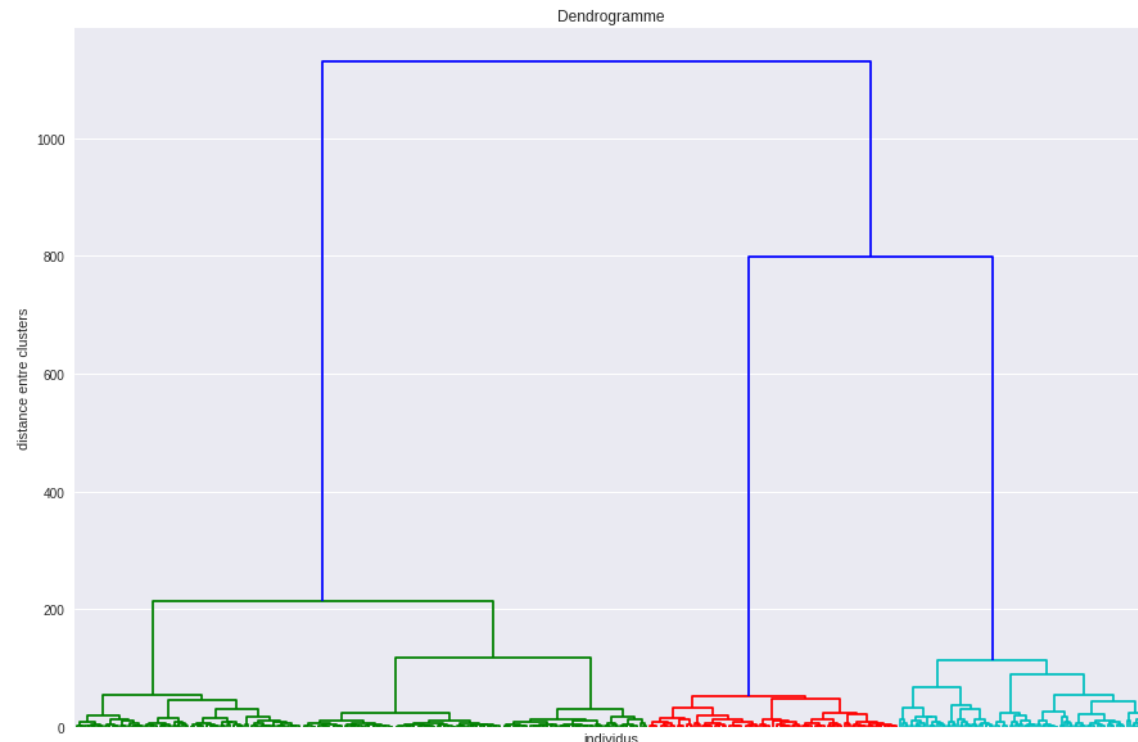
1. **Agglomerative:** Initially, each object is considered to be its own cluster. According to a particular procedure, the clusters are then merged step by step until a single cluster remains. At the end of the cluster merging process, a cluster containing all the elements will be formed.
2. **Divisive:** Initially, all objects are considered in a single cluster. Then the division process is performed step by step until each object forms a different cluster.

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Hierarchical clustering: Graphical representation

If we have a very large dataset, the bottom of the dendrogram may be unreadable.

⇒ Solution: Show and focus on the top part of the tree.

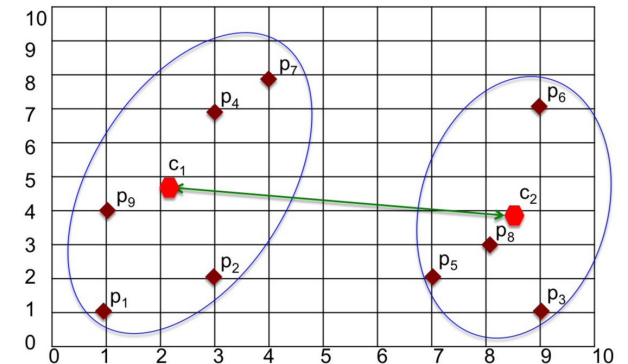
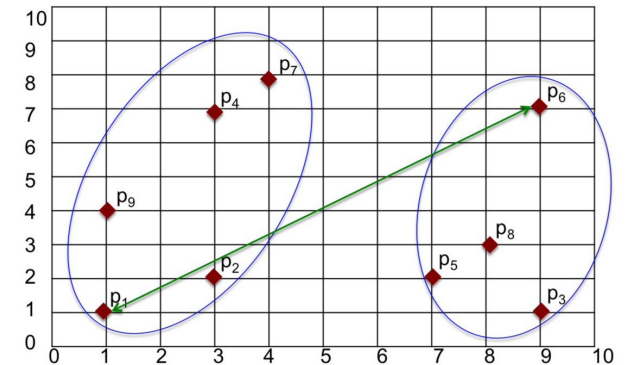
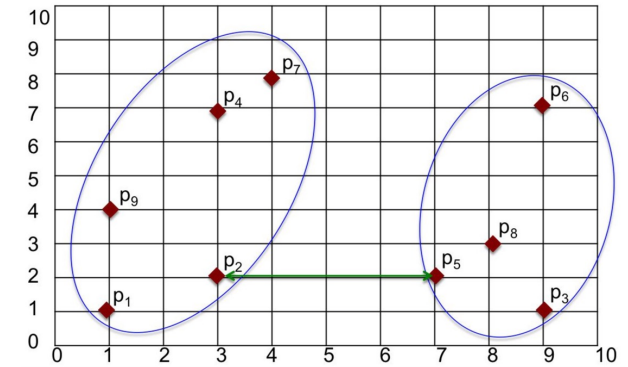


3. Clustering (unsupervised classification)

Hierarchical clustering

How to compute the proximity (similarity) between two clusters?

- **Min (Single) Linkage:** The distance between 2 clusters is the minimum distance between points in those clusters.
- **Max (Complete) Linkage:** The distance between 2 clusters is the maximum distance between points in those clusters.
- **Average Linkage:** The distance between 2 clusters as the average pairwise distance among all pairs of points in the clusters.
- **Centroid Linkage:** The distance between 2 clusters is the distance between their centroids.



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Hierarchical clustering

How to compute the proximity (similarity) between two clusters?

⇒ **Ward Linkage**

The Ward approach analyzes the variance of the clusters rather than measuring distances directly, minimizing the variance between clusters.

It consists of grouping clusters so that the increase in inter-cluster variance is maximum.

This amounts to the same thing according to Huygens' theorem: The increase in intra-class inertia is minimum.

Ward's method is preferred more than others in clustering.

3. *Clustering (unsupervised classification)*

Hierarchical clustering

Properties of the algorithm

- Unlike k-means, hierarchical classification does not require choosing a number of clusters beforehand.
- Algorithm independent of the order of insertion of points.
- Complexity: $O(n^3)$.